

A Case Study Report *on*

Is it a star, a brown dwarf, or a planet?

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ABSTRACT

This case study was conducted based on the preloaded data of the target object, HN Peg B, Sonora-Bobcat model (Marley et al. 2021) and Ackerman and Marley Cloudy model (Ackerman & Marley, 2001), via Google drive and Colab notebooks. The atmospheric models, bobcat and cloudy, predict the best fit parameters of the target object, T_{eff} (Kelvin), radius (R_{jup} – jupiter radius), metallicity/sedimentary efficiency, $\log g$ (cm/s^2), goodness-of-fit statistic to be 1300, 0.7996, 0.0, 3.4914, 7.3738 and 1300, 0.7838, 8.0, 4.4997, 5.1455 respectively. The model with the lowest goodness-of-fit statistic, cloudy, was used for further analysis.

Secondly, the evolutionary model was utilised to obtain the mass range, 10- 22 M_{jup} and radius range, 1.020 – 1.181 R_{jup} , of the target object, which was then be compared to the previously obtained data for discrepancies. Finally, after comparison with the published papers (Faherty et al. 2016), the target object was determined to be a planet and not a brown dwarf.

1. INTRODUCTION

A Celestial object can be defined as a planet, a brown dwarf or a star either by mass or formation (Burningham, 2024). It is a brown dwarf by mass, if it has a mass between 72 M_{Jup} (Jupiter mass) and 13 M_{Jup} (deuterium burning limit approximately) and by formation, if it has a mass limit 78 M_{Jup} (Chabrier et al. 2022).

Brown dwarfs, heat up at first, due to gravitational collapse and then cools down over time, resulting in temperature and luminosity to depend not only on mass but also on age. Similarly, radius of the brown dwarfs also follow a peculiar trend, it peaks at 4 M_{Jup}, but reduces as mass increase and decrease. In other words, radius of the brown dwarfs increases with mass and peaks at 4 M_{Jup}, after which it falls with increase in mass. Planets are formed from proto-planetary disks, leftover materials from star birth, either through core-accretion, solid particles come together to form terrestrial planets, or gravitational collapse, gas in the disk collapses due to dynamical instability thus forming gas giants (Burningham, 2024).

Sonora-Bobcat model, *named after the desert spanning the south western United States and northern Mexico* (Marley et al. 2021) has its own atmosphere model, Bobcat, as well as independent evolutionary models making it self consistent. Ackerman and Marley Cloudy model shows the scattering property of atmosphere and the effects of clouds on thermal emission. The parameters used in this case study are efficient temperature, radius, gravity, metallicity, abundance ratio, sedimentary efficiency and goodness-of-fit statistic. Temperature, radius and gravity are self-explanatory.

Metallicity (M/H) is a measure of how much metals exist in a celestial object and metals include any element heavier than Hydrogen and Helium. It is *the log of ratio of abundance of metals relative to Hydrogen, relative to value of sun* (Burningham, 2024). Additionally, younger stars have more metallicity than old stars because the universe is constantly enriched with metals through stellar nucleosynthesis. Abundance ratio (C/O) gives us information about the proportion of elements in the celestial object without solar reference. Sedimentary efficiency determines the balance between vertical mixing and gravitational settling of cloud particles (Ackerman & Marley, 2001). The higher the F_{sed}, the thinner the atmosphere will be as the large particles settle down due to gravity. Goodness-of-fit statistic refers to how much the data fits the model and is generally preferred to be low, denoting a better fit.

2. WORKING

In this case study, four Colab notebooks that makes the analysis a lot easier are used. Before starting the analysis, the correct directory path is set, in this case, my google drive and necessary libraries needed for the python code are imported. Apart from that, the given data, with the file name HN_Peg_b_Spex_Spectrum containing wavelength, flux and flux errors of the target object HN Peg B, placed in the google drive folder during set up is loaded and a flux vs wavelength graph is plotted.

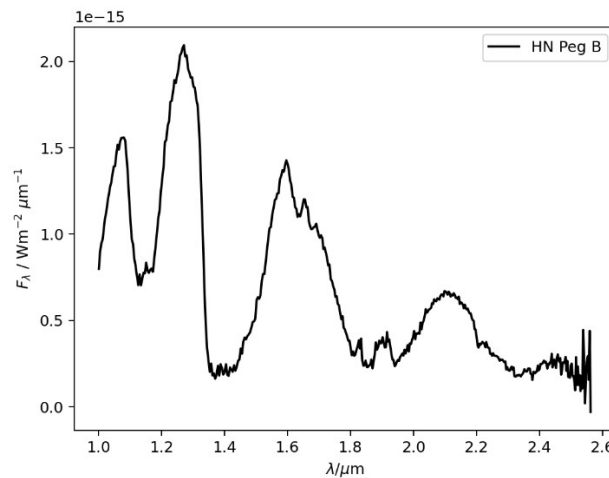


Figure 2.1. Wavelength vs flux graph for HN Peg B

In the upcoming part of the report, sample graphs for both the models, bobcat and cloudy, are drawn, followed by their respective best fits and the one with the most appropriate fit is selected for further analysis.

Let's proceed with Sonora Bobcat model. In this model, file names contain the parameters used to predict the spectra. For instance, in “*sp_t1000g1000nc_m+0.5_R3K.dat*”, sp refers to speclist.txt, the list of model spectra that came with the set up, t denotes the efficient temperature T_{eff} , g, gravity in m/s^2 , nc, no cloud and m, metallicity. All the models have data that were obtained from outside of the atmosphere, hence, it needs to be scaled before using with the data. Scale factor is the radius squared divided by distance squared (r^2/d^2). Then the models are convolved into the same resolution as the data for a meaningful approach [conv_uniform_R function is used to make both data and model have the same wavelength and other parameters, so it can fit properly].

After convolving the models to the same resolution, 500, with specific teffs, 1000, 800, 600, 400, sample graph was plotted (Figure 2.2.), that shows different efficient temperatures (1000, 800, 600, 400) at the same gravity (1000) and metallicity (0.0).

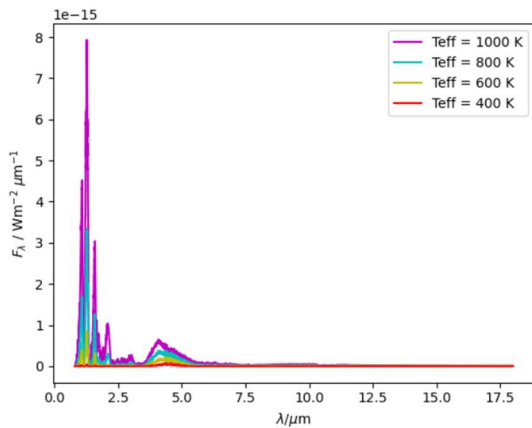


Figure 2.2. Sonora-bobcat model sample graph (Marley et al. 2021)

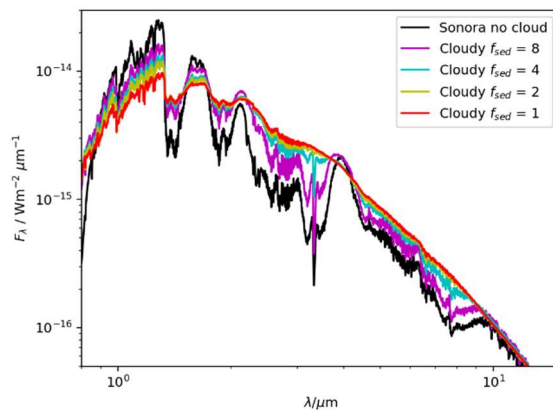


Figure 2.3. Cloudy model sample graph (Ackerman & Marley, 2001)

Advancing with the cloudy model, a similar approach is followed. Similar not the same because, the parameter for the cloudy model differs. The file name is ‘t1500g1000f1_m0.0_co1.0_std.spec_R3K.dat’, where t, g and m denotes the same, f indicates the sedimentary efficiency (Fsed) and co, the abundance ratio . It is also important to know that in this model m and co cannot be changed and are fixed at 0.0 and 0.0 respectively. Data with Fsed 1, 2, 4, 8 are loaded, scaled, with the scaling factor (r^2/d^2), convolved and then plotted into a sample graph Figure 2.3., which also has a no-cloud model included at the same t (1500) and g (1000).

Now, best fit among available goodness-of-fit statistic, in other words, the lowest goodness-of-fit statistic, is sought. A function that tells how much goodness-of-fit statistic and scaling factor is required is set up. After finding the distance of the target object from published papers and feeding it to the code along with the target object data and the preferred atmospheric model, we get the results in the tabular form (panda data frame used). The code is set up in such a way that the scaling factor is utilised to give out the radius in Rjup (Jupiter radius).

The target age, based on the chromospheric activity (Gaidos, 1998; Rocha-Pinto et al. 2004) and the debris disk (Bryden et al. 2006; Beichman et al. 2006) found around HN Peg A, is 0.3 ± 0.2 (Luhman et al. 2007). The target

distance adopted is *Hipparcos trigonometric distance*, 18.4 ± 0.3 parsec (Perryman et al. 1997).

Goodness-of-fit statistic, the lower the better. Hence to find the best fit, the lowest goodness-of-fit statistic among the given data for each model is attained and the most appropriate fit is decided. As per the data (Figure 2.4. and Figure 2.5.), cloudy model is the most appropriate fit than bobcat model. Hereon, Sonora-Bobcat model will be used for any further calculations.

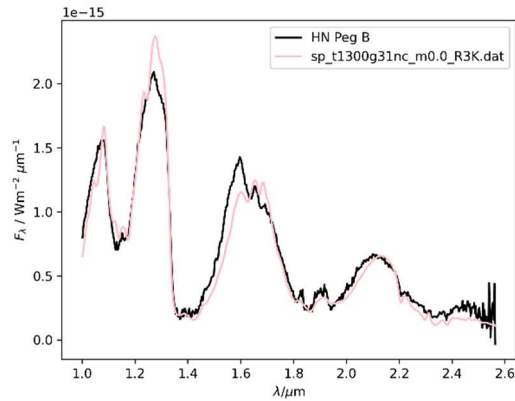


Figure 2.4. Best fit Bobcat model (Marley et al. 2021)

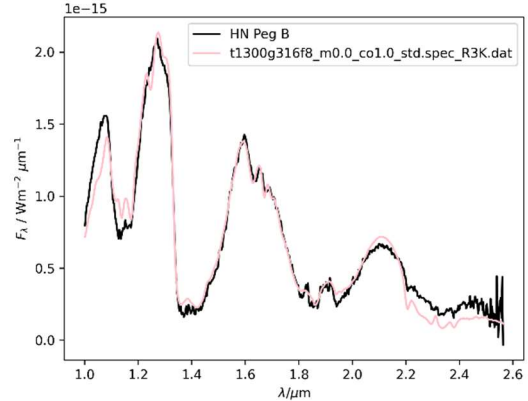


Figure 2.5. Best fit Cloudy model (Ackerman & Marley, 2001)

The best fit models, apart from giving the graphical data also gives out tabular data which is described in the Table 2.1 and Table 2.2.

Table 2.1. Sonora-Bobcat best fit model data

Parameters	Values
Teff (K)	1300
Log g (cm/s^2)	3.491362
Metallicity	0.0
Goodness-of-fit statistic	7.37376
Radius (R _{jup})	0.799582

Table 2.2. Cloudy best fit model data

Parameters	Values
Teff (K)	1300
Log g (cm/s^2)	4.499687
Sedimentary efficiency	8.0
Goodness-of-fit statistic	5.145506

Radius (R _{jup})	0.783811
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Before moving further, the luminosity of the target object is calculated, to find the mass range from evolutionary model. Luminosity of the target object is found using the formula $M_{bol} - M_{bol \odot} = -2.5 \log \frac{L}{L_0}$, where $M_{bol \odot}$ is the bolometric magnitude of sun of value 4.75, M_{bol} is the bolometric magnitude of the target, sum of the observed absolute magnitude M_{ks} and Bolometric correction, BC_{ks} (Filippazzo et al. 2015), $\log \frac{L}{L_0}$ is our required luminosity as the graph is plotted with the log instead of just luminosity.

Using the Bolometric correction BC_{ks} , 2.93759 ± 0.126 (Appendix A), the observed absolute magnitude M_{ks} , is estimated to be 14.3059 ± 0.2620 , with the distance modulus formula, $m_k - M_{ks} = 5 \log D - 5$, where m_k refers to the apparent magnitude in k band, 15.63 ± 0.249 (Skrutskie et al. 2006) and D, distance of the target object, 18.4 ± 0.3 pc. Further, the Bolometric magnitude M_{bol} is determined to be 17.2435 ± 0.0845 , resulting in the luminosity $\log \frac{L}{L_0}$, being -4.9974 ± 0.0338 . The values used for finding luminosity are described clearly in Table 2.3.

Table 2.3. Parameters for finding luminosity

Parameters	Values
$M_{bol \odot}$	4.75
M_{bol}	17.2435 ± 0.0845
M_{ks}	14.3059 ± 0.2620
BC_{ks}	2.93759 ± 0.126
m_k	15.63 ± 0.249
D (parsec)	18.4 ± 0.3
$\log L/L_0$	-4.9974 ± 0.0338

Continuing with the evolutionary models, data with solar metallicity of 0.0 is read into panda data frame. Among the many data, the array of data of masses that we are most interested in, is opted and plotted into a luminosity vs age graph (Figure 2.6).

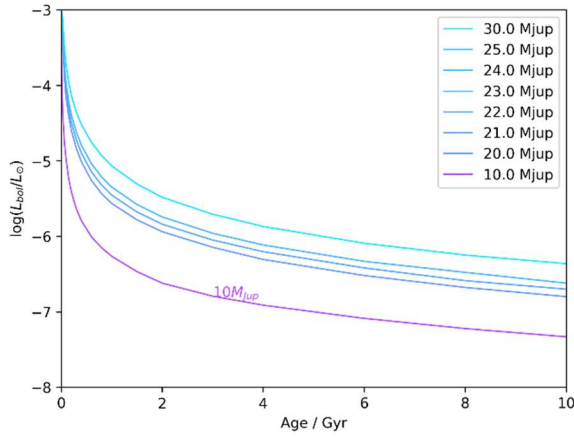


Figure 2.6. Luminosity vs age graph evolutionary model (Marley et al. 2021)

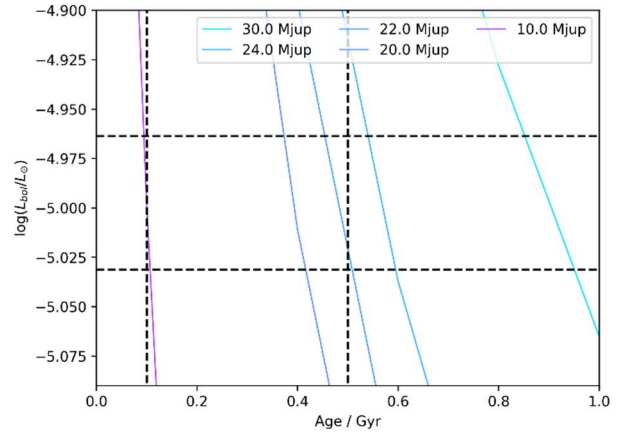


Figure 2.7. Determining mass range (Marley et al. 2021)

In the evolutionary model graph, horizontal and vertical lines are plotted, Figure 2.7, at luminosity (-5.0312 and -4.9636) and target age (0.1 and 0.5) respectively, thus filtering out the mass range 10 – 22 M_{Jup} . Next, radius vs age graph (Figure 2.8) is drawn to obtain radius from the evolutionary model. By drawing vertical lines from x axis at the target age (0.1 and 0.5), the radius range within the age limit is determined to be 1.020 to 1.181 R_{Jup} . Figure 2.9 depicts how this was done.

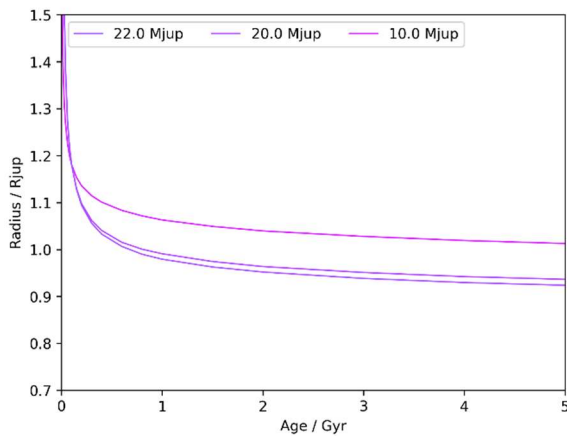


Figure 2.8. Radius vs Age graph evolutionary model (Marley et al. 2021)

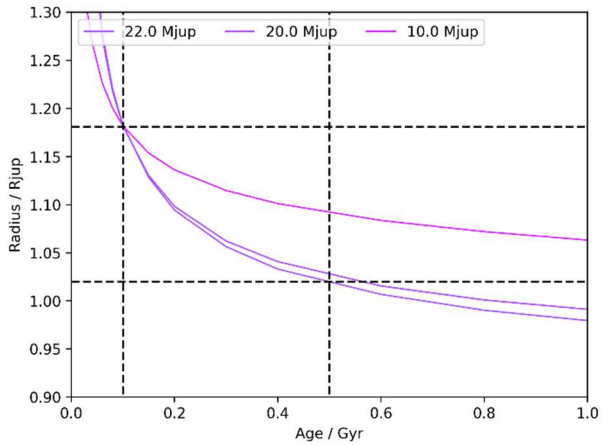


Figure 2.9. Determining radius range (Marley et al. 2021)

For comparison purpose, mass and T_{eff} are calculated using the formulae, $g = GM/R^2$, where M is the mass (meters), R , radius (meters), G , gravitational constant ($m^3 Kg^{-1} s^{-2}$) and g , gravity (m/s^2) and $L = 4\pi R^2 \sigma T_{\text{eff}}^4$, with L , luminosity, R , radius (m), σ , Stefan Boltzmann constant ($Wm^{-2} K^{-4}$) and T_{eff} ,

efficient temperature (K) respectively. The values for mass and T_{eff} are calculated to be 7.8 M_{jup} and 955.04 K to 987.87 K accordingly.

3. COMPARISON

In this section, the discrepancies seen between the values obtained in the atmospheric models as well as between the atmospheric model and evolutionary model is detailed.

3.1. Most appropriate fit:

From the figures, figure 2.4. and figure 2.5, it is obvious for the naked eye that Cloudy model fits better to the data than bobcat graphically. This could mean that the target object, HN Peg B, has a cloudy atmosphere, rather than no cloud. F_{sed} value of 8, implies heavy cloud particles, meaning they settle down into the object due to its gravity, rather than floating up like typical earth clouds. Such high F_{sed} value also signifies that the target object behaves similar to cloud-free objects.

3.2 Values of the atmospheric model:

Table 3.1. shows the parameters obtained from both the atmospheric models, bobcat and cloudy, side by side, for comparison purposes.

Table 3.1. Comparison table for atmospheric models

Parameters	Bobcat	Cloudy
Scaling factor	7.3737	5.1455
T_{eff} (K)	1300	1300
$\text{Log } g$ (cm/s^2)	3.4913	4.4996
Radius (R_{jup})	0.7995	0.7838

The scaling factor value further solidifies the previous estimate – cloudy model is a better fit. It is apparent that lower the scaling factor value, the better the fit.

T_{eff} , for both the model is identical, maybe because the atmosphere of an object at F_{sed} 8 is akin to a no cloud, hence T_{eff} observed from outside the atmosphere doesn't face any obstructions, displaying the same value.

Radius in R_{jup} , is almost similar, in both models ($\sim 0.8 M_{\text{jup}}$) indicating that the target could either be a Brown dwarf or a massive exoplanet. Such a small radius could mean two things, (a) object is older (> 1 Gyr) with low mass

(10-15 Mjup or below), (b) object is young (< 100 Myr) with high mass (Marley et al. 2021). But the target object falls into neither category, steadying between them. Hence, it can be guesstimated that the target is intermediate aged (100 Myr – 1 Gyr). Proceeding with this, we can justify the low radius of the target (0.7838), intermediate aged objects have a radius of 0.8 – 1 Rjup.

Log g values in both the models vary significantly, hinting two different things. Starting with Bobcat model, log g of 3.4913, indicates low surface gravity, meaning the object is not fully contracted, has thicker clouds, large radius and probably young as the gravitational contraction is not strong enough to drive these factors. Conversely, in Cloudy model, log g of 4.4996 denotes high surface gravity, meaning the object would be more compressed, has thin atmosphere, small radius and older.

Since Cloudy has a better fit, the estimates from cloudy model are considered.

3.3. Atmospheric models vs Evolutionary models:

Table 3.2. depicts the differing parameters we got from various sources – Atmospheric or evolutionary model through direct (retrieval from graph) or indirect (calculations) methods. The blue shaded boxes are those obtained through calculations.

Table 3.2. *Atmospheric vs Evolutionary model*

Parameters	Atmospheric model	Evolutionary model
Teff (K)	1300	955.04 – 987.87
Radius (Rjup)	0.7838	1.181 – 1.020
Mass (Mjup)	7.8	10 – 22

Teff calculated by the atmospheric model, doesn't consider the green house effects of planets, is affected by the clouds and chemical composition of the object. The huge difference we see in the evolutionary model Teff data might be due to the cloud interference and the chemical composition of HN Peg B. Radius and mass attained in both the models, atmospheric and evolutionary, have huge differences, maybe due to the different methods they use to measure. While evolutionary model considers the formation of the object, atmospheric component is compromised and atmospheric model takes that into account, it doesn't consider the whole object into consideration, just the atmosphere.

4. CONCLUSION

Based on this case study and further studies into determination of celestial objects, I hereby conclude the target object to be a exoplanet with low mass ($<10 M_{\text{jup}}$) and radius ($< 1 R_{\text{jup}}$). Though there are many uncertainties and discrepancies in the data obtained, one thing we can eliminate is the possibility of HN Peg B being a older object with heavier mass. HN Peg B is concluded to be an intermediate aged object, 100 Myr to 1 Gyr. Taking into consideration, the data acquired and other facts, there are large chances for HN Peg B to be a large exoplanet than a small brown dwarf.

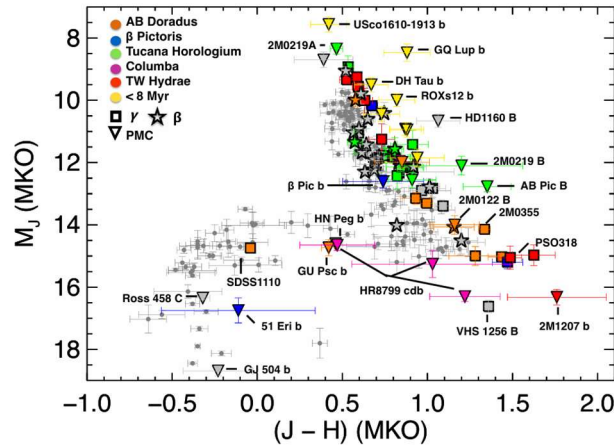


Figure 4.1. Colour magnitude diagram (Faherty et al. 2016)

Figure 4.1. describes the J-H versus M_j for brown dwarfs as well as directly imaged planetary mass companions (PMC) (Faherty et al. 2016). In this diagram, HN Peg B is considered as a PMC and not as a brown dwarf, but HN Peg B category is still not fixed due to various unknown factors, one of them being the lack of clear definition for brown dwarf and exoplanets. Determining the category of HN Peg B with undeniable facts is not possible with the current technology, but if done, it will be a turning point in categorisation.

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Appendix 1

Bolometric correction:

$$BC_{ks} = \sum_{i=0}^n c_i x^i$$

The target object is young with an estimated age of 0.1 to 0.5 Gyrs, hence the formula for young stars is used.

The values for c_i for a young celestial object are described in Table A1.1

Table A.1. BC_{ks} c_i and error values

Parameters	Values
c_0	-2.174
c_1	1.565
c_2	-1.759×10^{-1}
c_3	9.711×10^{-3}
c_4	-2.633×10^{-4}
c_5	2.742×10^{-6}
Error on BC_{ks}	0.126

Since our target is a T2.5 dwarf, the spectral type of T is 20 (Filippazzo et al. 2015), adding the 2.5 to 20, we get the spectral type of our target object, $x = 22.5$.